AI Summary of "Decreasing Universe Theory" (DUT)

ChatGPT:
https://jocaxx.github.io/DUniverse/Shorts/ChatGPT_DUT_Eng.pdf

DeepSeek:
https://jocaxx.github.io/DUniverse/Shorts/DeepSeek_DUT_Eng.pdf

Kimi:
https://jocaxx.github.io/DUniverse/Shorts/KIMI_DUT_ENG.pdf

Gemini:
https://jocaxx.github.io/DUniverse/Shorts/Gemini_DUT_Eng.pdf

Qwen:
https://jocaxx.github.io/DUniverse/Shorts/Qwen_DUT_Eng.pdf

NotebookLM:

https://jocaxx.github.io/DUniverse/Shorts/DUT_AI_1_ENG.pdf

https://jocaxx.github.io/DUniverse/Shorts/DUT_AI_2_ENG.pdf

Manus:
https://jocaxx.github.io/DUniverse/Shorts/Manus_Aval_Eng.pdf

The **"Decreasing Universe" Theory** (DUT)

Is an alternative cosmological proposal that seeks to explain phenomena such as the accelerated expansion of the Universe and galactic rotation curves

without invoking Dark Matter or Dark Energy.

Below is a detailed explanation of the pillars of this theory:

?? Core Idea: Gravitational Contraction of Space

The fundamental premise is bold: **the gravitational field continuously contracts space and everything within it** [[17]].

This means that:

* Space does *not* expand (as in the Big Bang/?CDM model).

* Instead, space (and matter) **shrinks** under gravitational influence.

* Crucially, **our measurement instruments** (rulers, atomic clocks, atoms themselves) are also slowly shrinking.

> ?? **The Ruler Analogy**: If your ruler shrinks by 7% every billion years [[25]], a fixed distance will appear *larger* when measured with it. Thus, distant galaxies appear to recede at an accelerated rate, but in reality, it is our "scale of measurement" that is changing.

?? How the Theory Explains Key Phenomena

1?? Apparent Accelerated Expansion (replacing Dark Energy)

In the standard model (?CDM), the acceleration of cosmic expansion is attributed to the mysterious **Dark Energy**.

In Barcellos' theory:

* Because our local standards of measurement (on Earth/in the Milky Way) are shrinking due to the gravitational field, cosmic distances appear to increase over time [[25]].

* This generates an **illusion of accelerated recession**, even deriving a version of **Hubble's Law** without requiring dark energy [[16]].

* The estimated contraction rate on Earth would be ~7% per billion years — sufficient to reproduce observational data [[25]].

2?? Galactic Rotation Curves (replacing Dark Matter)

The problem: stars at the edges of galaxies rotate faster than predicted by visible mass alone. The standard solution is to postulate **Dark Matter**.

Barcellos' solution:

* Gravitational contraction is not uniform: it is more intense near the galactic center and weaker at the edges [[17]].

* This differentially affects the wavelength of photons emitted by stars in different regions.

* When measuring velocities via redshift, this variation in contraction **mimics the effect of higher velocities**, reproducing rotation curves without invisible matter [[8]][[25]].

3?? Redshift (Cosmological Redshift)

In the conventional model, redshift is primarily a Doppler/expansion-of-space effect.

In the DU theory, redshift depends on four factors [[25]]:

Factor	Explanation
Photon travel time	The longer light travels, the more our "measurement standard" shrinks, increasing the measured redshift (dominant factor for high Z).
Local contraction of the emitter	The emitting galaxy also undergoes gravitational contraction, affecting the original wavelength.
Local contraction of the observer	Our current shrinkage alters the final measurement.
Relative mass	More massive galaxies contract more, which affects the observed redshift.

?? Testable Prediction: Key Difference from ?CDM

One of the most interesting contributions of the theory is a **falsifiable prediction** [[17]][[25]]:

> ?? **For galaxies at the same distance from Earth**:

> - **Decreasing Universe**: *More massive* galaxies should exhibit **lower redshift** (because they contract more locally, "compensating" part of the effect).

> - **?CDM**: *More massive* galaxies should exhibit **higher redshift** (due to additional gravitational redshift).

Barcellos claims to have compiled data favoring his prediction, challenging the standard model [[17]]. *It is worth noting that this analysis has not yet been widely replicated or validated by the mainstream cosmological community.*

?? Visual Summary

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STANDARD MODEL (?CDM):

Space expands ? Galaxies recede ? Redshift = expansion

+ Dark Energy (acceleration) + Dark Matter (galactic rotation)

BARCELLOS' THEORY (DUT):

Space CONTRACTS gravitationally ? Our rulers shrink ?

Galaxies APPEAR to recede ? Redshift = change in local scale

= No Dark Energy + No Dark Matter

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Yandex: **The "Decreasing Universe" theory** proposes that the gravitational field continuously contracts space and everything within it. This theory provides an alternative explanation for two key cosmological phenomena:

1. **The apparent accelerated expansion of galaxies**. According to this theory, the observed redshift is due to the contraction of space and measuring instruments, not to the galaxies moving away faster.
2. **The anomalous orbital velocities of stars**. The theory suggests that the contraction rate is higher closer to the galactic center, leading to a greater

reduction in emitted photon wavelengths, which mimics the effect of an increased Doppler shift for stars farther from the center.

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The theory does not directly derive from the effects of the equivalence principle and the Schwarzschild metric, but these effects illustrate that **gravitationally induced contraction is not an unreasonable concept**.

Some DUT articles:
https://jocaxx.github.io/DUniverse/Articles.pdf